Session 06 (AIML) - Assignment Questions

Q1. (Creating Lists)
Write a program to create the following lists using different methods:
a) A list of 5 integers directly.
b) An empty list using both [] and list().
c) A list containing mixed data types (int, string, float, boolean).
d) A list of 7 zeros using repetition (*).

Print all lists with clear labels and add comments explaining each creation method.

Q2. (Indexing and Negative Indexing)
Create a list: fruits = ["apple", "banana", "mango", "orange", "grape"]
Write code to print:

- First item
- Third item
- Last item (using negative index)
- Second last item (using negative index)

Take a number as input from the user and print the item at that index (handle basic cases only).

Q3. (Slicing Lists)
Given the list: numbers = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
Using slicing, print:
a) First 4 elements
b) Last 3 elements
c) Elements from index 2 to 7
d) Every second element
e) The entire list in reverse order

Add comments explaining slicing syntax.

Q4. (Modifying Lists)
Create a list: colors = ["red", "blue", "green", "yellow"]
Perform the following:

- Change "blue" to "purple" using indexing.
- Change the last item to "black".
  Print the list after each change.

Q5. (append() and insert())
Write a program that starts with an empty list.

- Use append() to add 5 different city names one by one.
- Use insert() to add one more city at position 2.
- Print the final list.

Take user input for at least 2 cities.

Q6. (remove(), pop(), and clear())
Create a list: items = [10, 20, 30, 20, 40, 50]
Perform and print after each operation:
a) Remove the first occurrence of 20 using remove().
b) Remove the item at index 3 using pop() and print the removed value.
c) Remove the last item using pop().
d) Clear the entire list using clear().

Add comments explaining the difference between remove() and pop().

Q7. (index() and count())
Create a list: scores = [85, 92, 78, 92, 65, 92, 88]

- Find and print the index of the first occurrence of 92.
- Count and print how many times 92 appears.
- Take a number as input from the user and check if it exists in the list. If yes, print its index and count.

Q8. (sort() and reverse())
Create a list: marks = [45, 78, 65, 90, 34, 82, 71]

- Sort the list in ascending order and print it.
- Sort the list in descending order and print it.
- Reverse the original list (without sorting) and print it.

Add comments on the difference between sort() and reverse().

Q9. (extend() vs append())
Create two lists:
list1 = [1, 2, 3]
list2 = [4, 5, 6]

Demonstrate:
a) Using extend() to add list2 to list1.
b) Using append() to add list2 to a copy of list1.

Print both results and explain the difference in comments.

Q10. (Mini Project - Combined Concepts)
Create a **Student Marks Manager** program using lists:

- Take 5 subject marks as input from the user and store them in a list.
- Print the list.
- Add one more subject mark using append().
- Print the highest and lowest marks (using built-in functions).
- Sort the marks in descending order.
- Calculate and print the average marks.
- Use len() to show total number of subjects.

Use proper comments, meaningful variable names, and control structures where needed.